New paleomagnetic age constraints on the Decaturville impact structure and Weaubleau structure along the 38th parallel in Missouri (North America)

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[1] Eight circular structures along the 38th parallel in the mid-continent United States are hypothesized to have formed from multiple impacts. Only two of these structures have been confirmed as impact features and they are not well dated. In this study we use paleomagnetic data to constrain the timing of the poorly dated Decaturville impact structure (Missouri) and the Weaubleau feature (Missouri), interpreted to be an impact structure, to evaluate the possibility of a serial impact event. A post depositional, mid Permian pole derived from impact-related breccias at Decaturville, as well as crosscutting relationships, show a Pennsylvanian-mid Permian age for the structure. A post-tilting Late Mississippian pole from deformed strata at Weaubleau indicates that the deformation event can be no younger than Late Mississippian. Our results demonstrate that the Decaturville and Weaubleau structures are of different ages and could not have been caused by a serial impact event. **Citation:** Elmore, R. D., and S. Dulin (2007), New paleomagnetic age constraints on the Decaturville impact structure and Weaubleau structure along the 38th parallel in Missouri (North America), *Geophys. Res. Lett.*, *34*, L13308, doi:10.1029/2007GL030113.

1. Introduction

[2] Many ancient impact structures are not well dated [*Koeberl and Anderson*, 1996] yet accurate geochronologic information is important in order to determine possible relationships with other geologic events (e.g., extinctions), to assess the possibility of serial impacts, and to study the frequency and periodicity of impacts on Earth. The rock record in North America contains many examples of structures interpreted to be impact in origin. Eight of these features lie along the 38th parallel in the mid-continent United States (Figure 1) [*Rampino and Volk*, 1996; *Melosh*, 1998; *Offield and Pohn*, 1979; *Evans et al.*, 2003]. The alignment of these features has led to the hypothesis that they could represent a serial impact similar to that observed on Jupiter during the impact of Comet Shoemaker-Levy 9 [*Rampino and Volk*, 1996]. This hypothesis, however, has been disputed [*Melosh*, 1998; *Luczaj*, 1998; *Rampino*, 1999; *Glikson*, 1999; *Koeberl and Reimold*, 1999; *Luczaj*, 1999] because only two of these structures have so far been identified as impact structures (Decaturville and Crooked Creek) based on the presence of shock metamorphic effects in the rocks, and also because the structures are not well dated.

[3] We report results of a modified paleomagnetic conglomerate test on impact breccias that show the poorly dated (Pennsylvanian-Cretaceous) [*Offield and Pohn*, 1979] Decaturville impact event occurred during the Pennsylvanian to mid Permian. We also use the results to investigate the origin of the impact-related breccias. In addition, the results are compared with data from the Weaubleau feature, interpreted to be an impact structure [*Evans et al.*, 2003], in order to test if the structures could have been caused by a serial impact.

2. Geologic Setting

[4] The Decaturville structure is a known 5.5 km diameter impact structure in Missouri (Figure 1) displaying shock metamorphic features, such as shatter cones [*Offield and Pohn*, 1979]. Ordovician rocks were near or at the surface during the impact, and there has been little deposition in the area since, which makes determination of the age based on stratigraphy difficult. Fission track dates on Precambrian schist at the center of the structure did not provide a reliable date for the age of the impact [*Offield and Pohn*, 1979]. *Offield and Pohn* [1979] suggested a Pennsylvanian to Cretaceous age range, whereas *Rampino* [1999] suggested a Silurian to Pennsylvanian age range for the structure.

[5] Underlying the Decaturville structure is a 540 m-thick sequence of Cambrian through Silurian stratified rocks overlying Precambrian crystalline basement [*Offield and Pohn*, 1979]. Accessible outcrops of impact related deposits within the structure are sparse, partly because most of the structure is on land with an uncooperative landowner. Polymict and monomcit breccias are common in the structure [*Offield and Pohn*, 1979]. Polymict breccias found outside of the central area are commonly associated with impact-related faults.

[6] A massive polymict breccia with no discernable dip and tan to gray dolomite beds and dolomitic breccias in the Ordovician Jefferson City Dolomite (JCD) were sampled at one road cut along Missouri Highway 5 in the southeastern quadrant of the structure (Figure 1). Samples were collected from clasts of the JCD and older units (Cambrian Davis Formation, Potosi Dolomite and Ordovician Roubidoux and Gasconade dolomites) in the polymict breccia (DV sites, Table 1) at the south end of the outcrop. The clasts sampled consisted of fine to medium-grained dolomites whereas the breccia matrix (site DV 17) is composed of dolomite, small clasts, and minor quartz. Ten sites were also collected from monomict breccia and dolomite beds in the JCD 2–50 meters

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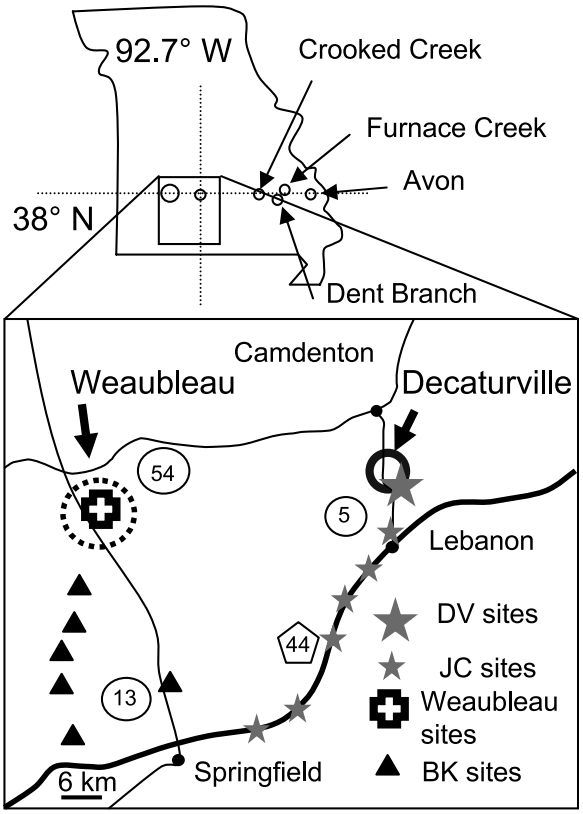


Figure 1. Map showing the locations of the Decaturville and Weaubleau structures and the sampling sites, as well as a map that shows the structures interpreted to be impacts along the 38th parallel in Missouri. Numbers correspond to road numbers.

north of the polymict breccia. These beds are faulted and contain small (<1–3 m) folds. Unbrecciated medium crystalline dolomites in the JCD (JC sites, Table 1) over 10 km outside the structure (Figure 1) were also sampled.

3. Methods

[7] A gasoline-powered portable drill was used to collect surface cores which were oriented with an inclinometer and a Brunton compass. Cores were cut to standard lengths (2.2 cm). All samples collected for paleomagnetic measurements were subjected to standard progressive thermal demagnetization procedures to isolate the characteristic remanent magnetization (ChRM) in a 2G Enterprises cryogenic magnetometer (Model 755-1.65) with DC squids. The results of a modified conglomerate test [*Hoblitt and Kellog*, 1979] were used to assess the timing and origin of the magnetization relative to deposition of the breccia at Decaturville. The conglomerate test is modified in that, with the exception of the JCD, the host rocks for most of the clasts were not sampled. Pole positions from the data were compared to the North American apparent polar wander path (APWP) [*Van der Voo*, 1993] to evaluate the timing of magnetizations. Conodont Alteration Index values (CAI) [*Epstein et al.*, 1977], geochemical data ($^{87}Sr/^{86}Sr$ analysis, University of Texas-Austin following the methods of *Gao et al.* [1992]), and petrographic (e.g., fluid inclusion) observations were performed to investigate the temperature history and origin of the breccia.

4. Results and Interpretations

[8] At low unblocking demagnetization temperatures (<250°C), the specimens contain a magnetization of north declination and steep positive inclination, which is interpreted as present day and viscous in origin. The polymict breccia clasts and matrix contain a ChRM that unblocks at higher temperatures (250–500°C; Figure 2a) with southeast declinations (D) and shallow inclinations (I) (D = 155.7°, I = −9.2°, k = 80.2 and $\alpha 95$ = 5.1°; n = means of 11/12 sites; Figure 2c and Table 1). The ChRM is also identified in alternating field demagnetization but the results were not used in the statistical analysis because the orthogonal projections displayed curved trajectories indicating overlapping components. The ChRM in the polymict breccia corresponds to a pole position of 50.0°N, 127.0°E (*dp* = 2.6°, *dm* = 5.1° [semi axes of the 95% ellipse of confidence around the pole]), which plots on the mid Permian part of the APWP (Figure 3). Because all of the clasts and the matrix have the same ChRM, the rocks were clearly remagnetized after deposition and subsequent to the impact.

[9] The clasts in the monomict breccia and the host JCD at the polymict site location contain a weak and unstable magnetization. As a result, directional data from these sites are not included in this paper. The JCD away from the impact, however, contains a ChRM (D = 142.1°, I = 4.3°, k = 112.7, $\alpha 95$ = 6.3°, n = 6 sites; Figures 2b–2c and Table 1) with a pole (36.8°N, 137.4°E (*dp* = 3.2°, *dm* = 6.3°) that is close to the Pennsylvanian part of the APWP (Figure 3) and is statistically distinct from the polymict breccia pole.

Table 1. Paleomagnetic Data From Decaturville[a]

Site	N/N_o	D_g, deg	I_g, deg	k	$\alpha 95$, deg
		Breccia			
DV 1	5/9	145.6	−26.2	38.7	12.5
DV 5	8/8	161.8	−9.8	83.8	6.1
DV 6	9/9	154	−4	51.9	7.2
DV 7	6/6	159	−10.6	71.8	8
DV 8	5/6	151.9	−10.1	93.3	8
DV 10	2/2	150.8	−16	103.5	24.8
DV 11	5/8	158.6	−1.3	24.9	15.6
DV 12	4/4	154.9	−10	52.1	12.8
DV 13	7/7	155.1	−13.3	79.4	6.8
DV 14	6/7	166.7	−6.5	32.7	11.9
DV 17	10/10	155.7	−9.3	118.8	4.5
DV 19	9/9	148.3	−0.3	191.4	3.7
Mean	11/12	155.7	−9.2	80.2	5.1
		JCD			
JC 1	12/12	151.7	3.1	1237.5	1.2
JC 2	11/11	146.3	−4.1	256	2.9
JC 3	7/7	142.1	6	203.1	4.2
JC 4	9/9	138.2	6.5	556.3	2.2
JC 5	11/11	138.6	10.4	740.9	1.7
JC 6	8/8	135.9	3.9	492.1	2.5
Mean	6/6	142.1	4.3	112.7	6.3

[a] N/N_o, number of specimens (or sites) with direction versus number demagnetized (used in mean calculation); Directions in geographic coordinates; units flat lying; D, declination; I, inclination; k, precision parameter; $\alpha 95$, cone of 95% confidence [*Fisher*, 1953]. Each DV site was collected in one breccia clast (2–9 specimens) or from breccia matrix (DV17).

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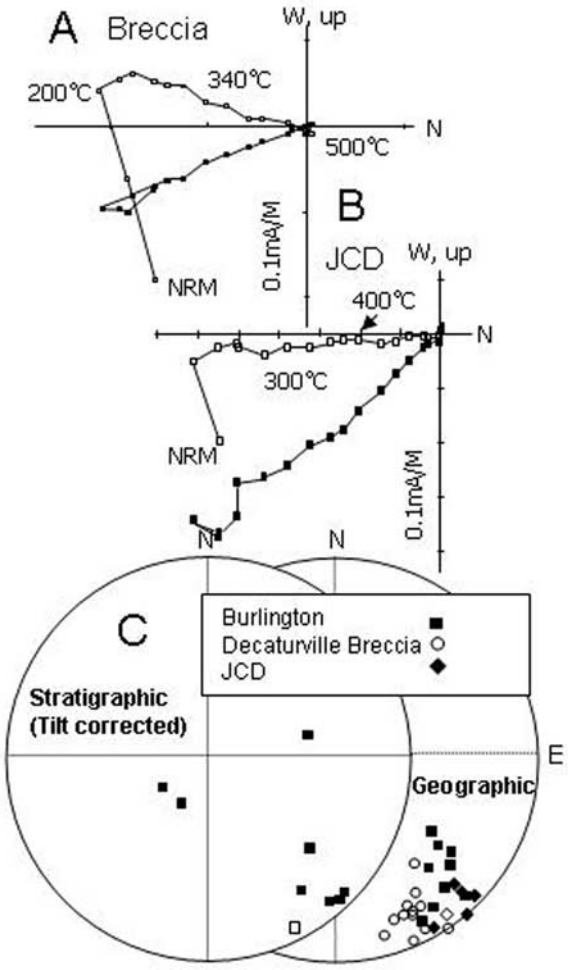


Figure 2. Orthogonal projections (no tilt correction) of representative specimens from the (a) polymict breccia and (b) the JCD outside of the structure. Horizontal projections: solid points; apparent vertical projections; open points. (c) Equal area projections showing site mean directions from the polymict breccia in the JCD at Decaturville (circles), the host JCD outside of the structure (diamonds), and from the Burlington-Keokuk Formation at the Weaubleau structure in geographic and stratigraphic coordinates (squares) [*Dulin and Elmore*, 2007]. Open symbols represent upper hemisphere projections; closed symbols are lower hemisphere projections.

The results of rock magnetic studies (e.g., acquisition and decay of an isothermal remanent magnetization) on samples from the polymict breccia and the JCD away from the structure are consistent with magnetite as the carrier of the remanence.

[10] Strontium isotopic analyses on representative samples of polymict breccia clasts (mean = 0.70831; standard deviation = ±0.00015; n = 4) show similar to depleted $^{87}Sr/^{86}Sr$ values compared to Ordovician seawater values (0.70875–0.70900) [*Qing et al.*, 1998], ruling out externally derived radiogenic fluids as an alteration source. The conodonts recovered from the clasts appear clear/slightly yellow, yielding a CAI of 1, which corresponds to heating below 100°C. Fluid inclusion analysis performed on polymict breccia clasts did not reveal useful inclusions, primarily because the dolomite that the clasts are composed of is too fine grained.

5. Discussion and Conclusions

[11] Paleomagnetic data from the modified conglomerate test results show that the polymict breccia at Decaturville contains a post-brecciation ChRM. This ChRM is mid Permian in age which indicates that the impact can be no younger than the mid Permian, regardless of the origin of the magnetization. *Zimmerman and Amstutz* [1972] and *Offield and Pohn* [1979] report that clasts of microbanded Mississippi Valley-type (MVT) sulfides occur in polymict breccias in the center of the structure. The clasts have sharp edges indicating that they did not crystallize in the breccia, and therefore, the brecciation occurred after the mineralization [*Zimmerman and Amstutz*, 1972]. Estimates for the age of similar mineralization in nearby MVT deposits range from Early Pennsylvanian to Triassic [*Symons and Sangster*, 1991; *Pannalal et al.*, 2004; *Symons et al.*, 2005]. This suggests that the impact structure could not be older than the Early Pennsylvanian, and along with the pole position obtained from the breccia, constrains the impact to the Pennsylvanian to mid Permian (Figure 3). The JCD outside the impact structure contains a ChRM that is older and unrelated to the ChRM at Decaturville.

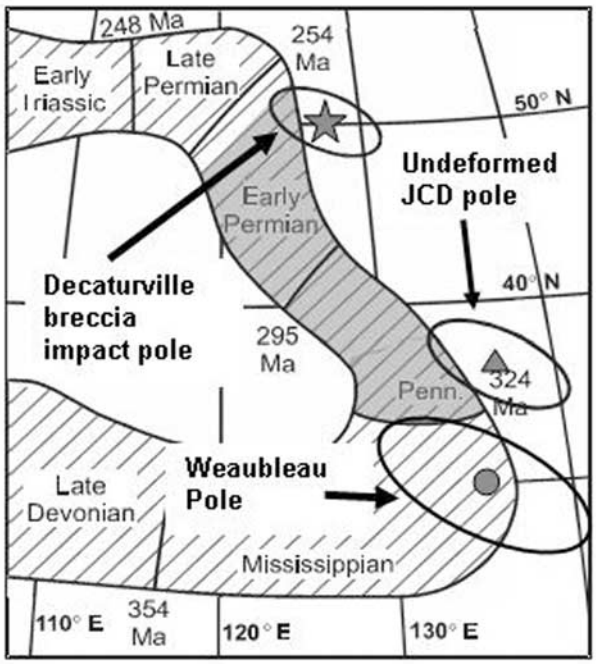


Figure 3. Late Paleozoic part of the APWP for North America (modified from *Symons et al.* [2005] and *Van der Voo* [1993]). The star indicates the pole position for the polymict breccia and matrix. The triangle represents the undeformed JCD. The solid circle represents the pole position for magnetization found at the Weaubleau structure [*Dulin and Elmore*, 2007]. The errors are represented by ellipses. The shaded part of the path corresponds to the age range for the Decaturville impact structure.

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[12] There are two likely possibilities for the origin of the ChRM in the breccia: a thermally induced remagnetization or a chemical remanent magnetization (CRM). The polymict breccia result is similar to many reported impact breccias that carry a thermal remagnetization [*Pohl and Soffel*, 1971; *Urrutia-Fucugauchi et al.*, 1994], in that the ChRM is post-depositional. The polymict breccia does not contain melt glass, melt, or recrystallization features that have been reported from other remagnetized breccias. Conodont analysis indicates temperatures less than 100°C which indicates that a thermal remagnetization is not likely. It should be noted that a low CAI for rocks associated with impacts has been reported from other impacts [*Jackson and Van der Voo*, 1986] and may not be an accurate record of temperature associated with impact. Heating associated with impact may not be sustained for sufficient time to alter conodonts, and, therefore, a low CAI value may not immediately rule out a thermal origin of the ChRM.

[13] A CRM, caused by a fluid flow event either related or unrelated to the impact, is also a possible mechanism for the ChRM. Other studies report that hydrothermal alteration associated with impact can lead to the acquisition of a CRM in impact-related rocks [*Pilkington and Grieve*, 1992]. The lack of elevated $^{87}Sr/^{86}Sr$ values in the polymict breccia samples rules out alteration by fluids with radiogenic strontium values. The CRM could be related to a fluid flow event associated with internally derived water. A chemical process is interpreted to be the most likely remagnetization mechanism for the ChRM in the polymict breccia.

[14] In a recently completed study, *Dulin and Elmore* [2007] report paleomagnetic results from the Weaubleau feature, interpreted to be an impact structure [*Rampino and Volk*, 1996; *Evans et al.*, 2003] or a cryptovolcanic feature [*Snyder and Gerdemann*, 1965; *Luczaj*, 1998]. Weaubleau, 80 km to the west of Decaturville (Figure 1), is a 19-km circular feature that exposes deformed Mississippian limestones. The deformation is interpreted to be related to the impact event [*Evans et al.*, 2003]. The absence of shatter cones and indexed planar deformation features in quartz grains leaves the origin of the structure in question [*Koeberl and Reimold*, 1999]. The age of the deformation is defined as younger than the deformed Osagean limestones and older than the overlying undeformed Pennsylvanian (Desmoinesian) strata.

[15] Specimens of the tilted Mississippian Burlington-Keokuk Formation limestone strata contain a ChRM that is unblocked by 475°C and resides in magnetite, based on rock magnetic studies [*Dulin and Elmore*, 2007]. The site means from deformed limestones group best in geographic rather than stratigraphic coordinates (Figure 2c). The results of a direction correction tilt test [*Enkin*, 2003] indicate a best estimate of −14.5% ± 16.9% untilting which overlaps the 0% unfolding interval, suggesting a post-deformational ChRM. The mean (geographic) of the ChRM (D = 138.9, I = 15.6 (k = 45.3, n = 9 sites, and $\alpha 95 = 7.7$) corresponds to a pole that lies at 30.3°N, 135.3°E ($dp = 4.1°$, $dm = 7.7°$) on the Late Mississippian part of the APWP (Figure 3). Samples from the Burlington-Keokuk Formation outside the structure (BK sites, Figure 1) are dominated by a present day magnetization and the ChRM is apparently localized within the deformation feature [*Dulin and Elmore*, 2007].

[16] The pole for the ChRM constrains the deformation to be Late Mississippian in age, which is more exact than the stratigraphic age constraint (Osagean to Desmoinesian). There are several possibilities for the origin of the ChRM. A shock remanent magnetization is pre-deformational and is, therefore, ruled out. A thermoviscous remagnetization can probably be ruled out because maximum unblocking temperatures (∼475°C) are too high considering that the rocks were only heated to low temperatures (<100°C for a CAI of 1) based on experimental evidence [*Kent*, 1985]. A post-depositional remanent magnetization, where the magnetic grains undergo passive rotation in fluid filled pores after deformation, is a possibility, particularly if the deformation occurred soon after deposition prior to lithification when the pores were still filled with water. A CRM acquisition by chemical alteration, either triggered by the deformation or a later event, is also possible. The limestones have radiogenic $^{87}Sr/^{86}Sr$ values, which suggest alteration by evolved fluids [*Dulin and Elmore*, 2007]. Fractures and faults produced by impact could have provided conduits for fluids that caused localized alteration although other deformation processes (i.e., cryptovolcanic) could also produce conduits for fluids.

[17] In summary, the modified conglomerate test on the breccia at Decaturville provides a significantly improved age range for the Decaturville impact structure (Pennsylvanian to mid Permian versus Pennsylvanian to Cretaceous). Tilt test results from the Weaubleau structure also provide a modest improvement in the age range (Late Mississippian versus Late Mississippian-Desmoinesian) for the deformation there. The age ranges do not overlap, which rules out a serial-impact origin for the two features.

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